WINNING THE SPACE RACE

with Data Science

Falcon 9 First-Stage Landing Prediction

IBM Applied Data Science Capstone Project

90	66.7%	83	83.33%
launch records	successful landings	engineered features	test accuracy

Executive Summary

- Goal: predict whether Falcon 9 first-stage recovery will be successful.
- Data combined public SpaceX API records with Wikipedia launch-history data.
- Workflow covered wrangling, EDA, SQL, Folium, Plotly Dash and supervised classification.
- 60 of 90 recorded launches produced successful first-stage landings (66.7%).
- Four tuned classifiers reached 83.33% accuracy on the 18-launch test split.

Introduction: Business Problem & Objective

- SpaceX's reusable first stage is a major contributor to lower launch costs.
- A competing launch provider can use historical data to estimate recovery likelihood before bidding.
- Research question: which mission, payload, orbit and launch-site characteristics are associated with successful recovery?
- Objective: build an end-to-end analytical and predictive workflow.

Methodology: Data Collection

- SpaceX REST API provided structured launch records and booster/mission metadata.
- Wikipedia scraping with BeautifulSoup supplied historical launch and landing information.
- JSON/API responses were normalized into Pandas DataFrames.
- Sources were aligned around launch date, booster, payload, orbit, site and outcome.

Methodology: Data Wrangling & Feature Engineering

- LandingPad was the major missing-value field (~28.9%).
- Created Class: 1 = successful landing, 0 = unsuccessful/no successful recovery.
- Applied one-hot encoding to categorical predictors.
- Final feature matrix contained 83 columns.
- Standard dataset: 90 records.

Methodology: Exploratory Data Analysis

- Used Pandas, Matplotlib and Seaborn for univariate and bivariate exploration.
- Compared landing outcomes across launch sites, orbit types, payload mass and flight number.
- Computed yearly landing-success rates.
- Used SQL for grouped counts, sums and mission-outcome analysis.

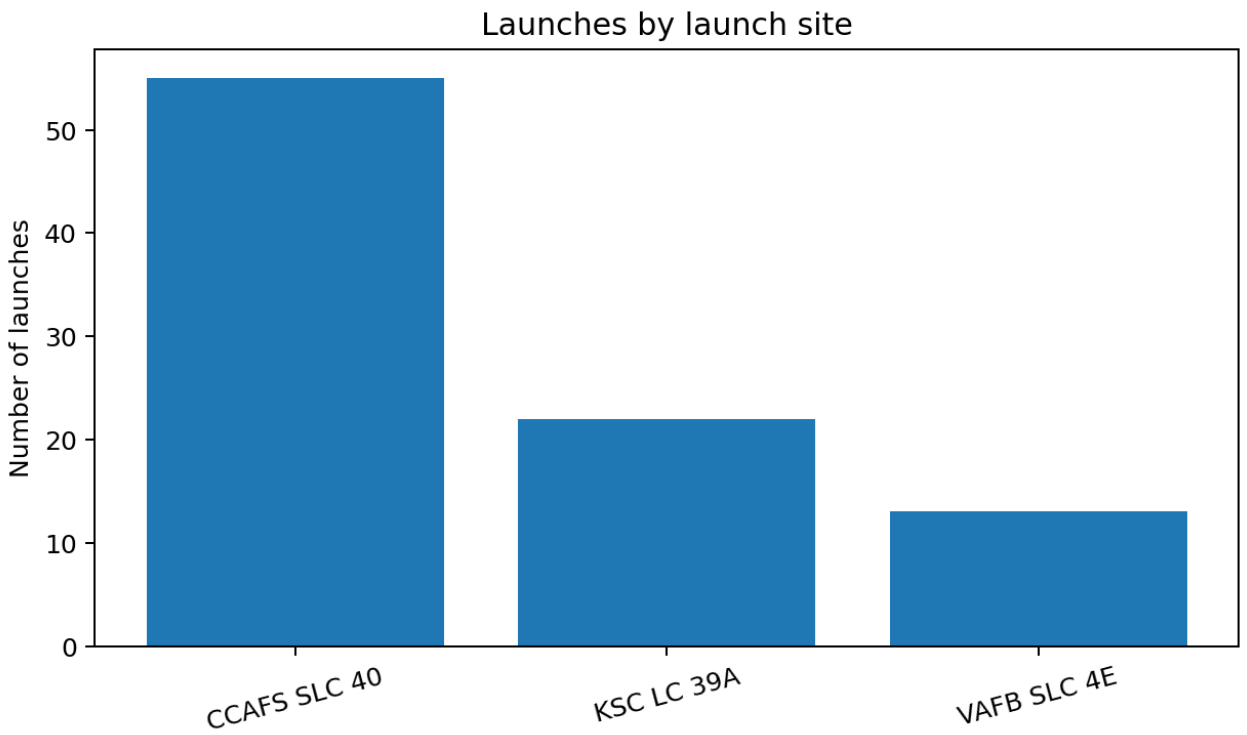
Methodology: Interactive Visual Analytics

- Folium mapped launch-site coordinates and landing outcomes.
- MarkerCluster organized nearby launch markers.
- Plotly Dash provided an interactive launch-site dropdown and payload range slider.
- Callbacks updated success summaries and payload/outcome views.

Methodology: Predictive Analysis

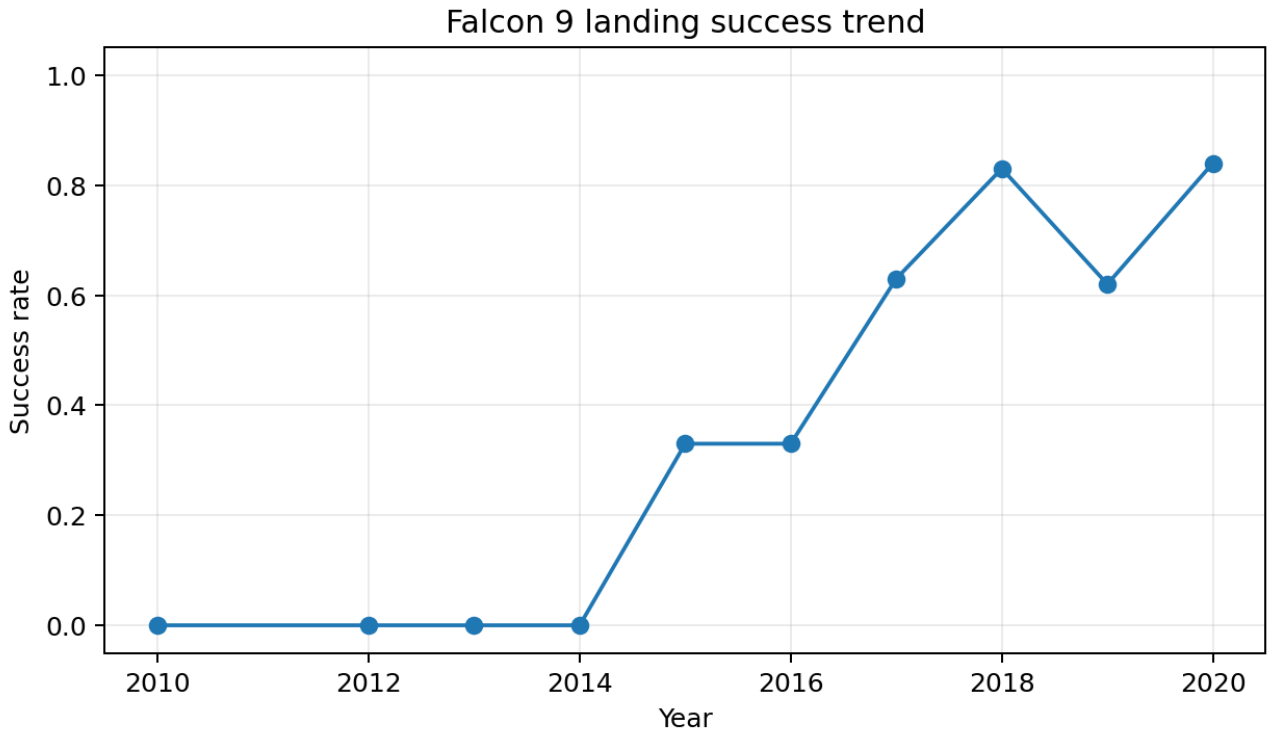
- Standardized the engineered feature matrix.
- 80/20 train-test split: 72 training records and 18 test records.
- GridSearchCV tuned model hyperparameters.
- Evaluated Logistic Regression, SVM, Decision Tree and KNN.

Results: Launch-Site Utilization



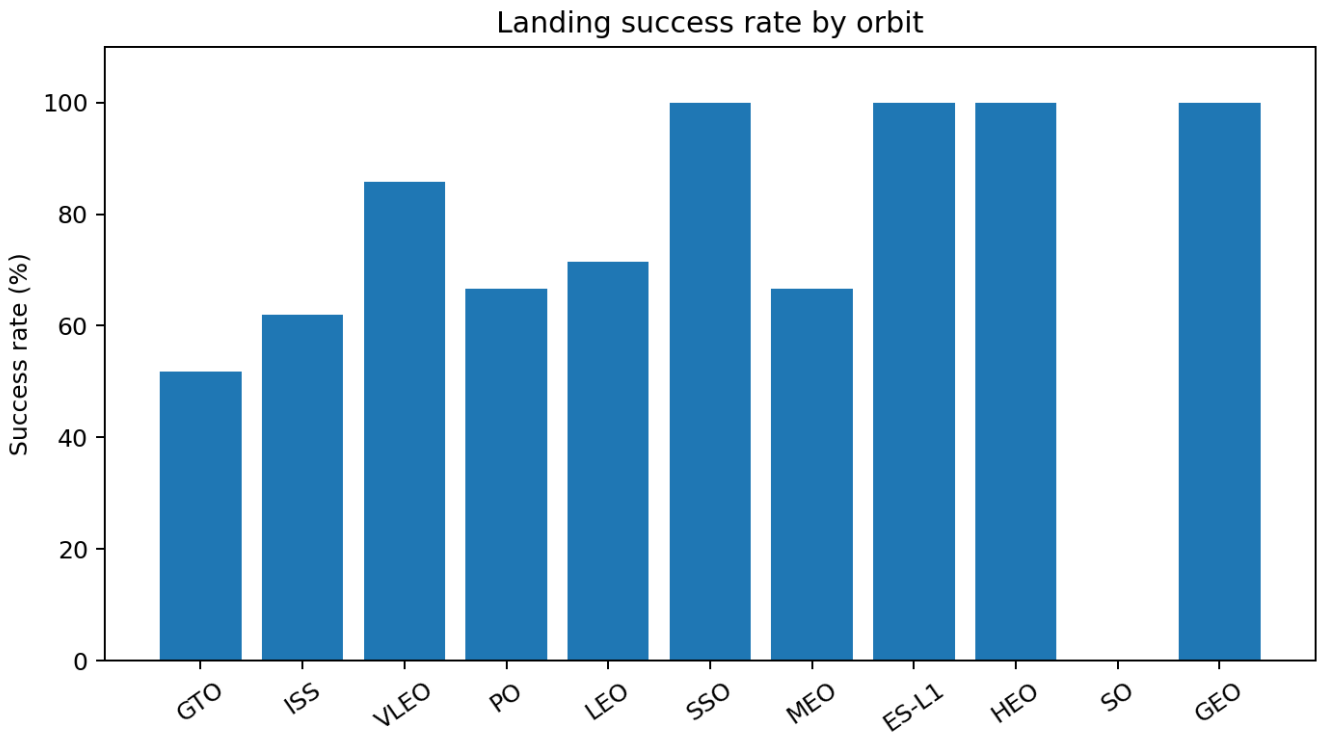
- CCAFS SLC 40: 55 launches.
- KSC LC 39A: 22 launches.
- VAFB SLC 4E: 13 launches.
- Later flight numbers show a visible shift toward successful recovery.

Results: Landing Success Improved Over Time



- Early flights had essentially no successful first-stage recovery.
- Success improved substantially from the mid-2010s onward.
- By 2019–2020, annual success rates were around the 80–90% range.

Results: Orbit Type Matters



- GTO: 27 missions and ~51.9% landing success.
- VLEO: ~85.7% success.
- ES-L1, GEO, HEO and SSO: 100% in this small sample.
- Orbit sample sizes are uneven; 100% rates should be interpreted cautiously.

Results: EDA with SQL

- 4 raw launch-site names; 3 cleaned operational sites.
- CCAFS SLC 40: 55 launches.
- NASA (CRS) total payload mass: 45,596 kg.
- Average F9 v1.1 payload mass: ~2,928 kg.
- Maximum payload mass: 15,600 kg.
- First successful ground-pad landing: 22 Dec 2015.

Results: Folium Interactive Map

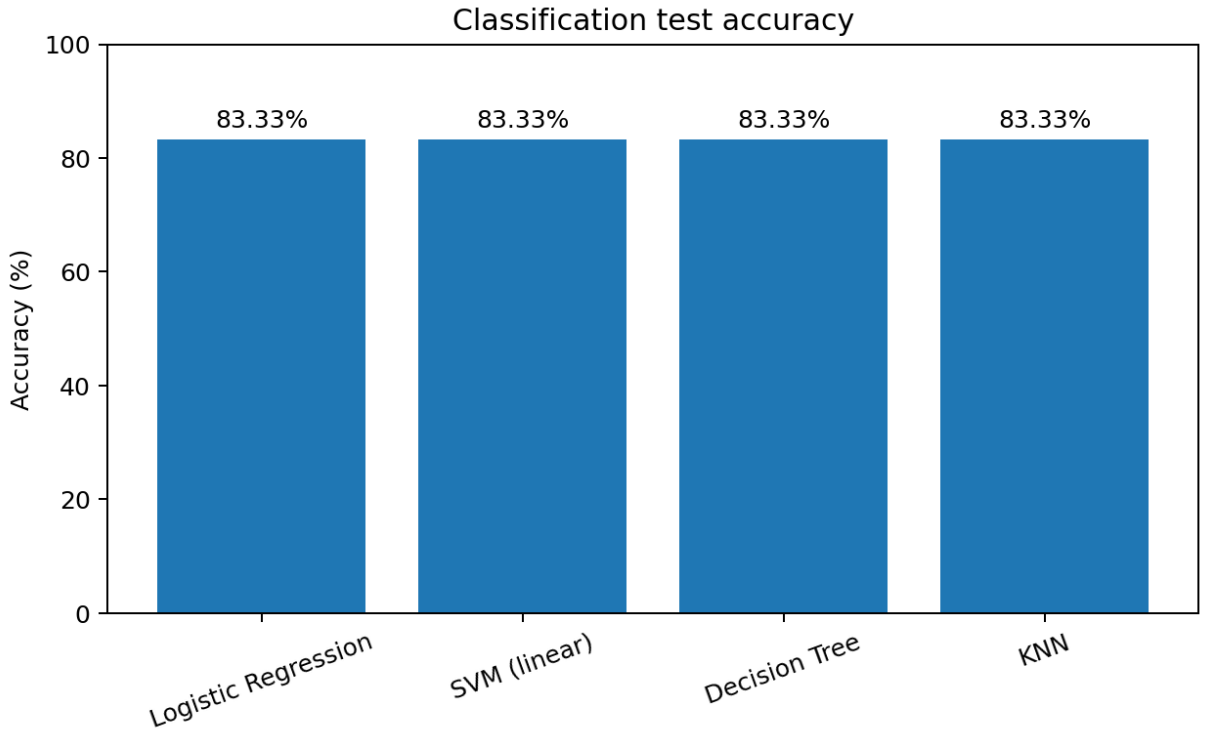
- Launch facilities are concentrated on U.S. coasts.
- CCAFS/KSC are clustered on Florida's east coast; VAFB is on California's coast.
- The map links geography to operational outcomes.
- The notebook also calculated proximity distances.

Results: Plotly Dash Dashboard



- Interactive launch-site dropdown changes the success summary.
- Payload slider filters the payload/outcome scatter plot.
- KSC LC-39A shows the strongest site-level success rate (~76.9%).
- Dashboard converts static EDA into interactive decision support.

Results: Predictive Classification



- All four classifiers reached 83.33% on the 18-record test split.
- SVM's selected kernel in the course workflow was linear.
- 83.33% = 15 correct predictions out of 18.
- Small test size means accuracy is sensitive to individual records.

Results: Confusion Matrix Interpretation

- 12 successful landings correctly identified.
- 3 unsuccessful cases correctly identified.
- 3 false positives predicted as successful.
- 0 false negatives in the commonly reproduced test split.

TN 3	**FP 3**
FN 0	**TP 12**

Innovative Insights: What the Data Suggests

- Operational learning is the clearest signal: landing success rises strongly with flight number and year.
- Launch site and orbit interact; GTO missions are materially harder to recover in this sample.
- Payload mass alone is not sufficient; orbit and booster generation matter.
- 83.33% test accuracy is promising but not definitive with only 18 held-out launches.
- More recent data would reduce uncertainty.

Conclusion

- Complete workflow: API/web collection → wrangling → EDA/SQL → Folium/Dash → ML.
- Overall landing success: 66.7%.
- KSC LC-39A showed the strongest site-level recovery rate.
- All four classifiers reached 83.33% on the standard test split.
- Next step: validate on newer, unseen missions.

Appendix: Deliverables & Reproducibility

- Replace the GitHub placeholder with your own repository URL.
- Include all notebooks and the Dash app.
- Final filename: Data Science Capstone Project Report.pdf
- Use the IBM/Coursera capstone template as the submission format.

Appendix: Key Sources

- IBM Skills Network capstone instructions/template.
- IBM Skills Network standard SpaceX datasets and guided notebooks.
- Public SpaceX API and Wikipedia launch-history records.
- Reference analyses were used to reconstruct standard course results because the original student notebooks were unavailable.